Sales and Rank on Amazon.com

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[PRELIMINARY DRAFT: COMMENTS WELCOME]

Abstract

In this note we provide a method for converting sales rank data into sales quantity. We show how sales quantity data can be extracted from product inventories on Amazon for a subset of products. We collect a large dataset using this method and then generalize results from Chevalier & Goolsbee (2003) by estimating the fit of a Pareto distribution model of sales rank for a large number of categories. We provide the outputs necessary to convert rank to quantity for these categories as well as a method for conducting inference on quantities.

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1 Introduction

The study of online retail platforms has been growing rapidly in importance and is now seen as central for ongoing debates in antitrust and industrial organization as well as the economics of retail, digitization, marketing, and even macroeconomics. Despite the broad interest in these platforms the main firms have control over their data and are generally opaque and reluctant to share this data with researchers.[1] As a result, both researchers and policymakers need to rely on what information is publicly available to study these markets. This is both a challenge and an opportunity. Because online retail platforms are in some respects publicly visible researchers have the ability to observe in real-time how markets react to policy changes, demand shocks, product introductions, etc.

The most visible data is product prices. As a consequence price data from online retailers has been widely used across many fields including industrial organization, which has studied the elasticity of online sales (Chevalier and Goolsbee (2003)) and magnitude of search costs online (Baye et al. (2006)), the consumer welfare impact of online retail (Brynjolfsson et al. (2003)) and online reviews (Waldfogel and Reimers (2020), Hollenbeck (2018)), the market for fake reviews (He, Hollenbeck, & Proserpio (2020)), the antitrust implications of pricing when sellers use algorithms (Mackay and Brown (2020)), the antitrust implications of platforms introducing their own competing products (Zhu and Liu (2018)) and many other issues. In macroeconomics online prices have been used to construct price indices to track inflation (Cavallo (2018), Cavallo (2017)) with implications for monetary policy.

These studies all have a significant shortcoming which is that while price data is abundant and available rapidly there is no corresponding quantity data. The primary analytical tools of economics rely on the existence of both price and quantity data, particularly for measuring demand curves, price elasticities, market shares and market definition, weighting price indices, and measuring the total size and scope of online retail. Measuring sales quantities

[1] An exception is eBay, who have allowed internal researchers to have access to data and has consequently received a disproportionate amount of attention in the academic literature relative to the larger and more traditional retail platform Amazon. See, for instance, Dinerstein et al. (2018), Einav et al. (2018)

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are also important for studying these platforms themselves, for example the effect of consumer ratings on sales, the effects of sponsored listings or other advertising, the importance of ranking algorithms, and so on.

This paper provides a general method for converting sales rank to sales quantities that can be used by researchers studying online retail platforms. We also collect sales data for a large number of products over a long timeframe and present results that researchers can use to directly convert sales rank to sales quantity. A few papers have made this conversion previously, in particular Chevalier and Goolsbee (2003) and Brynjolfsson et al. (2003), who study online price elasticities and consumer welfare, respectively. In both cases the authors use data on bestselling books as their setting. They both take advantage of the empirical regularity that sales distributions tend to follow a Pareto distribution and this allows for a straightforward estimation of the relationship between rank and sales quantity, provided data on sales exists. Chevalier and Goolsbee (2003) and Brynjolfsson et al. (2003) estimate this relationship for bestselling books using external data on book sales. The estimates from Chevalier and Goolsbee (2003) have been relied on by many subsequent studies interested in converting sales rank to sales, including [cite].

We generalize this approach by collecting sales data and estimating this relationship for a wide array of categories, not just for books and not just for the very high end of the sales rank distribution. Our first contribution is to show a method for collecting sales quantity data from Amazon.com. Next, we use this method to collect data for thousands of products over several months across a range of categories. We combine this with data on sales rank that we collect at a high frequency. We then estimate the model separately for each category and present detailed results that can be used by other researchers to convert sales rank data into sales. Finally, to demonstrate the empirical relevance of these results we calculate price elasticities for a range of products using data on price and sales rank and compare out-of-sample predictions of sales rank changes based on price changes.

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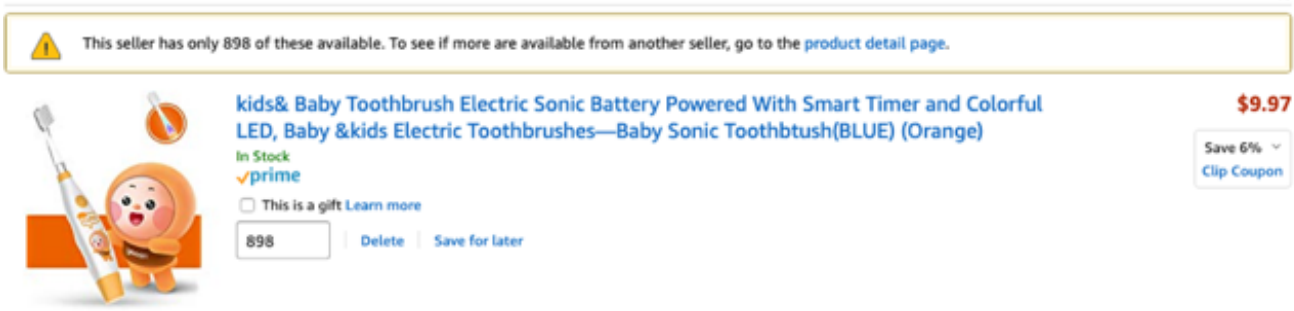

Figure 1: Inferring Amazon products stock

2 Overview of Data

We focus on Amazon.com because it is the largest online retailer in the U.S., capturing 49% of all e-commerce in 2018. Amazon displays on every product page a metric called Best Seller Rank.[2] Best Seller Rank, which we refer to throughout as rank, is a number that is assigned to each of the roughly 170 million products on Amazon. The number ranks products based on their rate of sales relative to other products in the same category.

Data Collection: To estimate the relationship between sales and sales rank, we first collect data on product sales for a wide range of products. To do so we exploit a feature of the Amazon website to infer the number of units of a product that are currently in stock. To observe this product inventory data, one must add to the Amazon cart increasingly larger numbers of units of the same products until the seller runs out of stock. At this point, Amazon will display an alert telling the buyer the total number of units available (see Figure 1). The highest number of units that can be added to an Amazon cart is 999 and so for products with inventories below 1000 we can observe the number of units currently in stock.

We employ research assistants to collect data using this method for a panel of products every 2 days. By observing inventories repeatedly over time, we can infer the rate of sales. In the next subsection we present summary statistics on inventories and sales. This method of

[2]Other online retailers present rank data as well, including Walmart.com and BarnesandNoble.com.

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measuring sales requires making several caveats: first, as already noted we can only observe sales for products with fewer than 1000 units in stock. But because sales rank is not a function of inventories this should not bias or distort the relationship between sales and sales rank. Second, restocking is unobserved. Therefore, if in a 2-day period a seller sells 50 units but re-stocks their inventory with 100 units, we will see the inventory increase and will not know how many sales took place. Third, some sellers limit the number of items a customer can purchase. In these cases we cannot observe the full inventory and our method does not provide useful data. These limits on purchases are relatively rare, however.

We first remove observations in which the inventory is 0 or at the upper limit of 999 or if the seller has placed a limit on the number of units that can be purchased. We then calculate the difference in inventories between each two day period. We remove any observations where the inventory increases over this period. We use the remaining data to calculate sales per day.

The second component of our data collection is to collect high-frequency data on sales rank for each of the products for which we have inventory sales data.[3] We also collect data on these products' prices, ratings and reviews, use of sponsored listings, and organic ranking in their keyword search page.

Summary Statistics Our data collection results in data from 1,293 products with valid observations on sales, i.e. those with repeated observations of valid inventory levels and no limits on purchases per customer. For these products we have 15,271 observations of sales rates. The products represent 17 major categories as defined by Amazon.

Table 5 presents summary information on product inventories, inferred sales, and sales ranks. The average inventory level among products with valid inventory levels (those between 1 and 998) is 241, with a median inventory of 150. Among products with valid observations of sales, the mean rate of sales per day is 9.8 with a median rate of 4 units sold per day.

[3]This data collection is described in more detail in He et al. (2020).

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	Mean	p5	Median	p95
Rank	80732.54	1914	39118.67	288794
log(Rank)	10.403	7.556	10.574	12.573
Inventory	241.316	4	150	791
Sales per day	9.859	.5	4	41.5

Table 1: Data summary

3 Empirical Method and Results

Our approach to recovering the relationship between sales quantities and sales rank follows Chevalier and Goolsbee (2003). The basic idea is to estimate the underlying distribution of sales that generate patterns in sales rank. The standard assumption is that sales follow a Pareto distribution. In this case, the probability that a value s exceeds the level S is the function $Pr(s > S) = (\frac{k}{S})^{\theta}$. For product sales ranks, this generates the relationship:

$$\frac{Rank - 1}{N} = (\frac{k}{Sales})^{\theta} \tag{1}$$

Here N is the total number of products in a category and k and θ are the parameters of the distribution that need to be recovered. These parameters can be estimated in a straightforward way. Taking logs of equation 1 yields

$$log(Rank - 1) = c - \theta log(Sales) \tag{2}$$

where c is a function of N, θ, and k. This specification directly generates a log-log regression specification to estimate the parameters of the distribution of sales.

We proceed to estimate equation 2 separately for each subcategory for which we have data. After which we provide a series of tests of model fit to justify the assumption of a Pareto distribution.

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	Mean	Median	p5	p95	N
R^2	.48	.493	.24	.791	17
c	10.859	11.128	7.831	12.012	17
θ	-.661	-.602	-1.007	-.465	17
N	807.294	577	112	2572	17

Table 2: Regression summary

3.1 Estimation Results

To estimate equation 2 we first take the average of daily sales data at the weekly level and use weekly observations of rank. We could alternatively estimate 2 at the daily level or some other level of aggregation. We tested different levels of aggregation and found quite stable estimates of $\hat{\theta}$ as well as overall model fit. Weekly data seem most useful for practical applications so we focus on that set of results. We exclude observations of greater than 100 sales in a given day as potential instances of mis-measured inventory data or outliers. This represents fewer than 1% of observations.

Table 1 summarize the result of these regressions and Table 2 shows the full set of regression results for each category in our data:

The number of observations in Table 1 refers to the number of category-level regressions we run. Despite the aggregation of the sales variable to the weekly level, the models fit fairly well with a median R^2 of .49. These R^2 values are lower than we would expect if sales rank was calculated purely as a function of the number of sales in the past 7 days and if those sales were always perfectly measured, indicating there is some loss to model fit from this simplification.

We get a mean and median of the key parameter θ of $-.66$ and $-.60$, respectively. To interpret this, note that in the Pareto distribution a θ less than 1 indicates that the probability of an event decreases less than linearly. A θ of .5 indicates that it decreases in the square root of its size, for instance. So our results suggest that for most categories sales quantities are fairly flat across a large range of sales ranks.

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Category		$\hat{\theta}$	$\hat{c}$	Observations	R^2	$\underline{c}$	$\overline{c}$	$\underline{\theta}$	$\overline{\theta}$
Arts, Craft, and Sewing	coef	-1.007***	11.43***	213	0.791	11.49	11.36	-0.926	-1.079
	se	(0.0470)	(0.0412)						
Automotive	coef	-0.985***	11.35***	250	0.595	11.16	11.55	-1.160	-0.811
	se	(0.113)	(0.125)						
Baby	coef	-0.465***	10.26***	469	0.388	10.19	10.33	-0.530	-0.401
	se	(0.0421)	(0.0466)						
Beauty and Personal Care	coef	-0.578***	11.49***	2,572	0.418	11.45	11.53	-0.609	-0.549
	se	(0.0187)	(0.0236)						
Camera, and Photo	coef	-0.713***	7.831***	166	0.562	8.042	7.614	-0.459	-0.959
	se	(0.154)	(0.132)						
Cellphones and Accesories	coef	-0.700***	10.50***	464	0.503	10.39	10.61	-0.775	-0.625
	se	(0.0488)	(0.0714)						
Clothing, Shoes, and Jewelry	coef	-0.605***	12.01***	233	0.439	11.98	12.05	-0.748	-0.453
	se	(0.0914)	(0.0217)						
Electronics	coef	-0.545***	9.642***	582	0.240	9.785	9.491	-0.424	-0.659
	se	(0.0703)	(0.0882)						
Health and Household	coef	-0.519***	11.51***	1,905	0.374	11.47	11.55	-0.550	-0.491
	se	(0.0191)	(0.0244)						
Home and Kitchen	coef	-0.684***	11.74***	2,209	0.412	11.71	11.78	-0.718	-0.651
	se	(0.0206)	(0.0213)						
Industrial and Scientific	coef	-0.994***	11.44***	112	0.675	10.73	12.15	-1.470	-0.516
	se	(0.296)	(0.441)						
Office Products	coef	-0.602***	10.79***	365	0.600	10.70	10.87	-0.673	-0.526
	se	(0.0463)	(0.0556)						
Patio, Lawn, and Garden	coef	-0.554***	10.70***	577	0.380	10.59	10.81	-0.625	-0.487
	se	(0.0445)	(0.0683)						
Pet Supplies	coef	-0.516***	10.72***	717	0.516	10.66	10.78	-0.569	-0.460
	se	(0.0331)	(0.0347)						
Sports, Outdoors	coef	-0.610***	11.13***	898	0.493	11.09	11.16	-0.644	-0.574
	se	(0.0215)	(0.0201)						
Tools and Home Improvement	coef	-0.580***	11.55***	830	0.508	11.49	11.60	-0.630	-0.525
	se	(0.0329)	(0.0353)						
Toys and Games	coef	-0.572***	10.51***	1,162	0.269	10.43	10.59	-0.629	-0.515
	se	(0.0362)	(0.0500)						

Table 3: Category-wise log-log regression results

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	Mean	Median	p5	p95	N
1day R^2	.476	.481	.234	.779	17
1day θ	-.657	-.596	-1.062	-.476	17
2day R^2	.48	.478	.237	.791	17
2day θ	-.66	-.603	-1.062	-.475	17
5day R^2	.484	.501	.244	.805	17
5day θ	-.665	-.613	-1.012	-.467	17
7day R^2	.48	.493	.24	.791	17
7day θ	-.661	-.602	-1.007	-.465	17
14day R^2	.469	.461	.238	.765	17
14day θ	-.671	-.586	-1.168	-.502	17
30day R^2	.479	.423	.285	.775	17
30day θ	-.742	-.64	-1.587	-.481	17
All + lags R^2	.901	.978	.489	1	17

Table 4: Regression summary of all specifications

3.2 Robustness and Specification Checks

In this subsection we perform robustness and specification checks to evaluate how well the Pareto assumption fits the data, how our results are affected by aggregation, and how our results vary over the range of possible sales ranks in the data.

First, as noted in the previous subsection the average R^2 is only .49. This is likely driven by the fact that the sales rank is actually a ranking of a weighted average of sales over different time periods, i.e. sales over the past hour, the past day, the past week, the past month, etc. Somewhat low R^2 also could result from a mis-specified model, measurement error, or other concerns. We therefore run a regression that includes logs of the rate of sales for different time periods as well as lagged values of those variables on the right hand side to evaluate how well this richer regression can predict rank. We include sales over the past 1, 2, 5, 7, 14, and 30 days and report the distribution of R^2 across categories in Table 4. The regression that includes all variables including lags performs substantially better at explaining variation in rank, with a median R^2 of .978, although we note that this regression does not result from a theoretical argument as the original regression follows from equation

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Category	q25	q50	q75
Arts, Craft, and Sewing	-0.863	-0.973	-1.093
Automotive	-0.959	-0.889	-0.901
Baby	-0.501	-0.518	-0.502
Beauty and Personal Care	-0.582	-0.591	-0.578
Camera, and Photo	-0.606	-0.690	-0.727
Cellphones and Accesories	-0.742	-0.734	-0.682
Clothing, Shoes, and Jewelry	-0.645	-0.706	-0.628
Electronics	-0.668	-0.753	-0.795
Health and Household	-0.542	-0.566	-0.585
Home and Kitchen	-0.703	-0.716	-0.659
Industrial and Scientific	-0.809	-0.724	-0.235
Office Products	-0.657	-0.704	-0.672
Patio, Lawn, and Garden	-0.563	-0.763	-0.692
Pet Supplies	-0.552	-0.558	-0.566
Sports, Outdoors	-0.581	-0.537	-0.614
Tools and Home Improvement	-0.643	-0.675	-0.652
Toys and Games	-0.549	-0.677	-0.747
Mean	-0.651	-0.692	-0.671

Table 5: Quantile Regression

1 and perhaps more important, is not invertible. Knowing rank does not allow one to back out quantity, in other words, only a weighted average of different quantities.

Given that we prefer to include only one quantity variable in the rank regression, it is not obvious which time aggregation to include and how sensitive our results are to this aggregation. Table R^2 shows how the mean parameter $\hat{\theta}$ and R^2 varies over different time aggregations. It shows that both the model fit and the estimated relationship between sales and rank change very little depending on this assumption. Weekly data is typically used in empirical applications so our main results all use the weekly specification.

Finally, we investigate how the relationship between rank and sales varies over the range of possible ranks in the data. It may be the case, for instance, that θ differs whether one is interested in bestselling products compared with products with few sales at the very bottom of sales ranks. We therefore run quantile regressions of equation 2 at the (25th, 50th, 75th) percentiles of the rank distribution. Table 5 shows the mean estimates of θ at each of these.

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Overall while there is some noise in the category-level estimates, the relationship is quite stable. The mean θ estimated at the 25th percentile of rank is -.65 and the mean θ estimated at the 75th percentile is -.67.

3.3 Inference

A key output of this research is the ability to convert readily available data on sales ranks to sales quantities. To do so, one must merely invert equation 2 and calculate

$$\hat{Q} = exp(\frac{\hat{c} - log(Rank - 1)}{\hat{\theta}}) \tag{3}$$

where $\hat{Q}$ is estimated weekly sales. To construct a valid confidence interval for $\hat{Q}$ one cannot simply plug in the confidence interval values for $(\hat{c}, \hat{\theta})$ because of the non-linearity of equation 3. Instead, we propose the following:

1. Draw a very large number of random draws from the estimated bivariate normal density of the two coefficients $(\hat{c}, \hat{\theta})$
2. For each draw construct the output of interest $\hat{Q}$, and call this distribution $\{\hat{Q}_s\}$
3. Find the 5th and 95th percentile values of $\{\hat{Q}_s\}$ and then save the (c_s, θ_s) associated with them.

These values can be used to construct confidence intervals for any $\hat{Q}$ and are reported in Table 4 for all categories.

3.4 Applications

4 Conclusion

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References

Baye, M., Morgan, J., and Scholten, P. (2006). Information, Search, and Price Dispersion. *Handbook in Economics and Information Systems*, 1.

Brynjolfsson, E., Hu, Y. J., and Smith, M. D. (2003). Consumer surplus in the digital economy: Estimating the value of increased product variety at online booksellers. *Management Science*, 49(11):1580–1596.

Cavallo, A. (2017). Are online and offline prices similar? evidence from large multi-channel retailers. *American Economic Review*, 107(1):283–303.

Cavallo, A. (2018). Scraped Data and Sticky Prices. *The Review of Economics and Statistics*, 100(1):105–119.

Chevalier, J. and Goolsbee, A. (2003). Measuring Prices and Price Competition Online: Amazon.com and BarnesandNoble.com. *Quantitative Marketing and Economics*, 1(2).

Dinerstein, M., Einav, L., Levin, J., and Sundaresan, N. (2018). Consumer price search and platform design in internet commerce. *American Economic Review*, 108(7):1820–59.

Einav, L., Farronato, C., Levin, J., and Sundaresan, N. (2018). Auctions versus Posted Prices in Online Markets. *Journal of Political Economy*, 126(1):178–215.

He, S., Hollenbeck, B., and Proserpio, D. (2020). The Market for Fake Reviews. *https://ssrn.com/abstract=3664992*.

Hollenbeck, B. (2018). Online reputation mechanisms and the decreasing value of chain affiliation. *Journal of Marketing Research*, 55(5):636–654.

Zhu, F. and Liu, Q. (2018). Competing with complementors: An empirical look at amazon.com. *Strategic Management Journal*, 39(10):2618–2642.

Electronic copy available at: https://ssrn.com/abstract=3728281